

System Requirements Documentation

Project Title: Clay'sCars.com

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Table of Contents

1. Customer Problem Statement and System Requirements
2. Functional Requirement Specification
3. System Sequence Diagram
4. Activity Diagram
5. User Interface Specification
6. Traceability Matrix
7. System Architecture and System Design
8. Algorithms and Data Structures
9. User Interface Design and Implementation
10. Design of Tests
11. Project Plan
12. References

1. Customer Problem Statement:

Residents in Madison, Indiana have limited ways to view available cars across local dealerships. Clay'sCars.com consolidates used and new vehicle listings into a single browser-accessible site with filtering options.

2. Functional Requirements:

- Filter cars by Make, Model, and Year
- Display real-time, clickable listings
- Navigate to dealer websites directly

3. System Sequence Diagram:

User -> UI Dropdown -> Filter Logic -> Display Listings -> Link to Dealer

4. Activity Diagram:

Start -> Select Filters -> Process Input -> Render Results -> User Action -> Exit

5. UI Specification:

- Three dropdowns: Make, Model, Year
- Dynamic filtered results
- Clean, minimal layout

6. Traceability Matrix:

Requirement	Implemented In
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FR1: Filter by make/model/year	JS filter logic
FR2: Display results cleanly	HTML results section
FR3: Link to dealers	Anchor tags in results

7. Architecture & Design:

- Single-page HTML app
- JavaScript handles filtering and data rendering
- No backend/database; data is embedded

8. Algorithms & Data Structures:

- JavaScript arrays and filter() logic for searching
- Set() for dynamic dropdown values

9. UI Design & Implementation:

- HTML/CSS layout
- Responsive design
- Logical element structure

10. Design of Tests:

- Manual UI testing for dropdown filter logic
- Tested expected outputs and empty input behavior

11. Project Plan:

Week 1: Research & Requirements

Week 2: Design & Data Collection

Week 3: HTML/JS Prototype

Week 4: Polishing & Packaging

12. References:

- Dealership websites in Madison, IN
- Mozilla Developer Network
- W3Schools JavaScript documentation